Statistical Signal Processing
Lecture 08:
Non-parametric estimation of PSD (spectral estimation)

Literature and Problems: See course schedule

Agenda (preliminary):

Introduction to spectral estimation

- **Theoretical calculation of PSD in two ways**
- **Practical constraints: finite signal length, random variations, noise**
- **Desired properties for PSD estimates wrt. bias, consistency, MSE, complexity**
- **SLIDE: Overview of methods for spectral estimation**
- **Brief DFT/FFT review (optional)**

The Periodogram estimate

- **Definition**
- **Properties: biased but asymptotically unbiased, not consistent, resolution**
- **DEMO: examples**

The Bartlett estimate

- **Definition using signal slicing and averaging of periodograms**
- **Properties: biased but asymptotically unbiased, consistent, resolution**
- **DEMO: examples**

The Welch estimate

- **Definition using signal slicing, overlapping blocks, windowing and averaging of periodograms**
- **Properties: biased but asymptotically unbiased, consistent, resolution**
- **DEMO: examples**

The Blackman-Tukey estimate

- **Definition using windowing of estimated autocorrelation function**
- **Properties: biased but asymptotically unbiased, consistent, resolution**
- **DEMO: examples**

SLIDE: Performance comparison of methods for spectral estimation

- **Variability**
- **Frequency resolution**
- **Figure of merit**
- **Computational complexity**

Brief introduction to non-stationary signals and the Short Time Fourier Transform (optional)